

1.2.4

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u \quad \text{has a true angular frequency } \omega$$

Assuming wave travelling diagonally $k_x = k_y = k$, continuous dispersion is:

$$\omega = c |k|_2 = c k \sqrt{2}$$

when simulating this on the grid, computer produces slightly flawed numerical freq $\tilde{\omega}$

Prove that under specific conditions, i.e. $c = \dots$, $\tilde{\omega} = c \sqrt{2} = \omega$.

1) Take eqn. 1.2 $u_{i,j}^{n+1} = c^2 (u_{i+1,j}^n + u_{i-1,j}^n - 2u_{i,j}^n) + u_{i,j}^{n-1}$ & plug it in 1.3 LHS

$$\frac{u_{i,j}^{n+1} - 2u_{i,j}^n + u_{i,j}^{n-1}}{\Delta t^2} = \text{RHS} \Rightarrow u_{i,j}^n \cdot e^{Ei} - 2e^{Ei} + \frac{u_{i,j}^n}{e^{Ei}} = E =$$

Factor out $u_{i,j}^n$: $u_{i,j}^n (e^{Ei} - 2 + e^{-Ei}) = u_{i,j}^n (e^{-\tilde{\omega} \Delta t} - 2 + e^{\tilde{\omega} \Delta t})$

Use Euler's formula: $e^{xi} + e^{-xi} = 2 \cos(x)$

Braket simplifies: $u_{i,j}^n (2 \cos(\tilde{\omega} \Delta t) - 2) = -2 u_{i,j}^n (1 - \cos(\tilde{\omega} \Delta t))$

Recap trigonometric identity $(1 - \cos(\theta)) = 2 \sin^2(\theta/2)$

$$-4 u_{i,j}^n \sin^2\left(\frac{\tilde{\omega} \Delta t}{2}\right)$$

2.) Repeat substitution for RHS:

$$\text{LHS} = \frac{u_{i,j}^{n+1} - 2u_{i,j}^n + u_{i,j}^{n-1}}{\Delta t^2} = c^2 \left(\frac{u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n}{\Delta x^2} + \frac{u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n}{\Delta y^2} \right)$$

$$\frac{-4 u_{i,j}^n \sin^2\left(\frac{\tilde{\omega} \Delta t}{2}\right)}{\Delta t^2} = c^2 \left(\frac{-4 u_{i,j}^n \sin^2\left(\frac{k \Delta x}{2}\right)}{h^2} + \frac{-4 u_{i,j}^n \sin^2\left(\frac{k \Delta y}{2}\right)}{h^2} \right) \cdot (-4 u_{i,j}^n) \quad h = \Delta x = \Delta y$$

* since $k_x = k_y = k$
x & y axis
will give
same result!

$$\sin^2\left(\frac{\tilde{\omega} \Delta t}{2}\right) = \frac{c^2 \Delta t^2}{h^2} \left(2 \sin^2\left(\frac{k h}{2}\right) \right)$$

$$\sin^2\left(\frac{\tilde{\omega} \Delta t}{2}\right) = 2 c^2 \sin^2\left(\frac{k h}{2}\right) \quad \leftarrow c = 1/\sqrt{2} ; c^2 = \frac{1}{2}$$

$\sin^2\left(\frac{\tilde{\omega} \Delta t}{2}\right) = \sin^2\left(\frac{k h}{2}\right)$ / if we assume small step size & equate the sine arguments directly

$$\frac{\tilde{\omega} \Delta t}{2} = \frac{k_y}{2} \quad / \cdot 2$$

$$\tilde{\omega} = \frac{k_y}{\Delta t}$$

← narrow angle CFL condition $\xi = c \frac{\Delta t}{\Delta h} = 1/\sqrt{2}$
 $\downarrow$
 $\frac{h}{\Delta t} = c\sqrt{2}$

$$\tilde{\omega} = c\sqrt{2}$$

numerical frequency perfectly matches physical frequency; ergo no dispersion error!

1.2.3

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \quad \leftarrow \text{insert } u(x, y, t) = e^{i(k_x x + k_y y - \omega t)}$$

$$\frac{\partial u}{\partial t} = \frac{\partial}{\partial t} (e^{i\Phi(x, y, t)}) = \frac{\partial (e^{i\Phi(x, y, t)})}{\partial z} \frac{dz}{dt} = e^{i\Phi} \cdot -i\omega = -i\omega e^{i\Phi} = -i\omega u$$

$$\frac{d^2 u}{dt^2} = \frac{\partial u}{\partial t} \frac{\partial}{\partial t} = i^2 \omega^2 u = -\omega^2 u$$

Spatial derivatives:

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x} (e^{i\Phi(x, y, t)}) = \frac{\partial (e^{i\Phi(x, y, t)})}{\partial z} \frac{\partial z}{\partial x} = e^{i\Phi} \cdot i k_x$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = i^2 k_x^2 u = -k_x^2 u$$

$$\frac{\partial^2 u}{\partial y^2} = -k_y^2 u \quad \text{follow for the y-derivatives}$$

→ substitute derived identities back into the wave eqn.

$$-\omega^2 u = c^2 (-k_x^2 u - k_y^2 u)$$

$$-\omega^2 \cancel{u} = -c^2 \cancel{u} (k_x^2 + k_y^2) \quad / \cdot u$$

$$+\omega^2 = +c^2 (k_x^2 + k_y^2) \quad (\sqrt{\quad})$$

$$\boxed{\omega = c \sqrt{k_x^2 + k_y^2}} \quad \leftarrow \text{continuous dispersion relation}$$